Reinforcement-Learning-Scheduled Latent Diffusion for Visual Counterfactual Explanations, and What Ground-Truth-Anchored Evaluation Reveals About Them

Anonymous CVPR submission

Paper ID ****

Abstract

Visual counterfactual explanation methods for medical image classifiers are conventionally scored by Counterfactual Validity (CV): the rate at which a generated image flips the classifier's decision. We make two contributions that pull in opposite directions, and argue the second is the more consequential. First, we present PPO-CF*, a system in which a PPO agent schedules the sampling procedure of a frozen, LoRA-adapted latent diffusion generator — selecting per-step noise strength, classifier- and preservation-guidance scales, refinement-pass count, guidance ramp-up point and stopping time — rather than applying one hand-tuned schedule to every image, with editing spatially confined to the lung field by a RePaint-style background lock. Second, we show that CV, computed as it usually is with the same classifier that guided generation, is close to uninformative about whether the edit is class-relevant at all. We introduce two measurements that do not share that circularity: a Lesion Hit Rate (LHR) anchored to radiologist-drawn opacity boxes, and a held-out transfer ratio that re-scores counterfactuals with classifiers that never participated in guidance. Under these, the picture changes substantially. On the RSNA Pneumonia Detection Challenge, our committed configuration attains a mean CV of* 0.838 *and a* $2.67\times$ *LHR lift over the annotation-area null (*$p = 4.2 \times 10^{-53}$*), yet held-out transfer remains in the adversarial regime on the most independent validator (*0.046*/*0.186 *against a* 0.15 *threshold). A five-arm study, each arm a policy trained from scratch and evaluated under an identical protocol, isolates ensemble guidance as the only tested countermeasure that helps (paired* $d = 0.482$*) and produces two confirmed negative results: an annealed sparsity reward fails at its own objective and costs lesion targeting (*$d = 0.512$*), and a Grad-CAM spatial constraint collapses the base task entirely (*$d = 1.205$*, CV* $0.838 \rightarrow 0.017$*). We argue that reporting CV without a ground-truth- or independence-anchored companion metric systematically overstates progress in this literature.*

1. Introduction

Classifiers deployed on chest radiographs are typically audited with attribution maps, which indicate *where* evidence is concentrated but not *what change* would alter the prediction. Visual counterfactual explanations close this gap: given a query image x with prediction $f(x)$ and a target class c, a generator produces x_c that (i) is scored at c by the classifier, (ii) remains a realistic radiograph, and (iii) differs from x only in a minimal, class-relevant region, preserving anatomy, foreign objects and patient identity.

Two observations motivate this work. The first is a modelling one. A fixed, hand-tuned guidance schedule is a poor fit across a heterogeneous test population: some images need only a gentle nudge across the decision boundary, others require aggressive multi-pass editing, and one schedule either under-edits the latter or de-realizes the former. We therefore frame per-step guidance control as a sequential decision problem and train a PPO agent to schedule it, image-adaptively, against a reward that trades classification progress against preservation and realism. Crucially the agent never touches the diffusion weights; it controls only the sampling procedure around a frozen generator, which keeps the action space small and the learned policy reusable at inference time on arbitrary query images.

The second observation is an evaluative one, and in retrospect the more important. The field's standard success criterion, Counterfactual Validity, is almost always computed with the same classifier whose gradients guided generation. It therefore cannot distinguish a genuine class-relevant edit from an adversarial perturbation tuned to one model. This is not a hypothetical concern in our setting: our own system reaches high CV while placing only a minority of its edit mass inside radiologist-drawn lesion annotations, and while transferring poorly to classifiers that did not participate in guidance. We consider it a reporting problem for the

subfield rather than a defect specific to our generator, and we address it with two companion measurements — one anchored to human annotation, one to model independence — that we recommend alongside CV.

Contributions.

1. PPO-CF: a PPO-scheduled sampling controller over a frozen, LoRA-adapted Stable Diffusion generator, with a hard segmentation-gated background lock guaranteeing that content outside the lung field reconstructs to the autoencoder's identity floor (§3).
2. An evaluation protocol that pairs CV with a ground-truth-anchored Lesion Hit Rate and a held-out transfer ratio, together with the split-, seeding- and routing-level controls a control-arm comparison requires in order to mean anything (§4).
3. A five-arm controlled study in which every arm is a policy trained from scratch and fully evaluated, yielding one confirmed improvement and two confirmed negative results, including the largest effect size we measure — against a mechanism designed to help (§5.5).

2. Related Work

Counterfactual explanation in medical imaging. Singla *et al.* [16] introduced a GAN-based counterfactual generator for chest radiographs together with the evaluation protocol — Counterfactual Validity, FID, and Foreign-Object Preservation — that has become the de facto benchmark, and which we adopt for comparability. Their FOP metric depends on a Fast R-CNN foreign-object detector trained on private annotations, which is not reproducible offline; §4.1 describes our model-free substitute. Subsequent work has largely retained the same classifier-scored notion of validity [1, 7, 8].

Diffusion-based counterfactuals. Classifier- and gradient-guided diffusion sampling [5, 10, 11] has displaced GAN-based approaches for editable generation, owing to more stable training and higher sample fidelity. We build on a LoRA-adapted [6] Stable Diffusion v1.5 [12] backbone with DDIM sampling [17] rather than training a diffusion model from scratch on a comparatively small radiograph corpus.

Spatially constrained editing. RePaint [9] and Blended Latent Diffusion [2] re-noise and re-composite the unedited region at every denoising step. We adapt this as a hard background lock: rather than trusting the guidance gradient to keep edits inside the lung field, the latent outside a dilated, feathered lung mask is forcibly reset at every step to the original image's forward-diffused latent.

RL and diffusion. Most RL work on diffusion fine-tunes the model *weights* against a reward (e.g. DDPO [3], reward-weighted fine-tuning). Treating the sampling hyperparameters as an RL action space around a frozen generator is comparatively unexplored, and is what we do here.

3. Method

3.1. Problem formulation

Given a frozen binary classifier $f : \mathcal{X} \to [0, 1]$ and a target class $c \in \{0, 1\}$, we seek x_c minimizing

$$\mathcal{L}(x_c) = \underbrace{\mathrm{BCE}(f(x_c), c)}_{\text{classification}} + \underbrace{\lambda_{\text{pres}}\, d(x, x_c)}_{\text{preservation}}, \qquad (1)$$

where d is a masked distance penalizing change inside the segmented lung field and change to a frozen object-detector embedding (a foreign-object proxy), and leaving change outside the lung field free — subject to the hard lock below, which removes that freedom entirely.

3.2. Latent generator and background lock

Images are encoded to a $4 \times 32 \times 32$ latent by a frozen SD1.5 VAE. A lightweight MLP maps $(c, \text{pooled latent})$ to a single cross-attention conditioning token, replacing SD1.5's text conditioning with a scalar class signal. Rank-8 LoRA adapters ($\alpha = 16$, dropout 0.05) are inserted into the UNet's `to_q`/`to_k`/`to_v`/`to_out.0` projections and trained on the standard ϵ-prediction objective over the training split, adapting the natural-image prior toward radiography without touching the VAE or the bulk of the UNet.

Guided DDIM sampling. Generation runs a partial, SDEdit-style forward-then-reverse DDIM trajectory whose length is a `strength` fraction of $T_{\text{diff}} = 50$ steps, with two gradient terms injected into the predicted $\hat{z}_0$ at each step:

$$\Delta z_0 = -w(i)\alpha_{\text{cls}} \nabla_{\hat{z}_0} \mathcal{L}_{\text{cls}} - w(i)\alpha_{\text{pres}} \nabla_{\hat{z}_0} \left(L_1^{\text{seg}} + \eta\, \mathrm{MSE}_{\text{obj}} \right), \qquad (2)$$

gated by a dilated, Gaussian-feathered lung mask M and a linear ramp $w(i)$ rising from 0 at a learned fraction of the trajectory, which avoids the discontinuity of a hard guidance switch.

Background lock. At every step the latent outside M is overwritten with the original latent forward-noised to the current timestep (clean at the final step). This is a hard spatial constraint independent of how the guidance gradient behaves: content outside the lung field is guaranteed to collapse to the VAE reconstruction floor rather than being repainted by the free SDEdit process, which is what an unconstrained high-`strength` schedule does to the entire thorax.

3.3. PPO-scheduled sampling

Rather than fixing `strength`, guidance scales and pass count, a PPO agent [14] emits a 9-dimensional continuous

Table 1. Action space. All nine dimensions are emitted per refinement step, so the schedule is adaptive both across images and within an episode.

	Controls	Range
a_0	DDIM strength	$[0.20, 0.70]$
a_1	classifier-guidance scale	$[0.5, 3.0]$
a_2	preservation-guidance scale	$[0.3, 1.5]$
a_3	stop signal τ_t	$[0, 1]$
a_4, a_5	extra refinement pass (Bernoulli)	$\{0, 1\}$
a_6	per-pass strength boost	$[0.85, 1.15]$
a_7	guidance bias (cls vs. pres)	$[0.70, 1.30]$
a_8	guidance ramp-up start fraction	$[0.1, 0.5]$

action at each of up to $T = 25$ refinement steps, mapped through sigmoid transforms to the controls in Table 1.

The observation concatenates the flattened original and current latents, the normalized step index, the current $|f(x_t) - c|$, the target c, and the flattened lung mask. Episodes terminate when $|f(x_t) - c| < \varepsilon_{\text{stop}} = 0.08$ or the budget is exhausted.

Reward. The per-step reward combines progress toward the target, a fidelity penalty against the original, a mask-weighted preservation penalty plus a foreign-object-embedding KL term, a per-step budget cost, and terminal bonuses for reaching validity and for stopping once valid, weighted as in the supplementary configuration.

Architecture. A shared $512 \to 256$ MLP trunk feeds a Gaussian policy head with per-dimension learnable log-std, a value head, and three auxiliary heads (feasibility, progress, stop) trained alongside the clipped PPO objective with GAE ($\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$).

3.4. Directional policies

The environment is built around a fixed target, so one policy is trained for n→p ($c = 1$) and another for p→n ($c = 0$), sharing the generator but specializing their schedules. This follows an asymmetry we observe throughout: synthesizing a plausible, spatially coherent opacity pattern in a clean lung field is a harder and more under-constrained generative problem than removing one, which is closer to an inpainting operation the pretrained prior already performs well.

4. Evaluation Protocol

4.1. Conventional metrics

Following [16], the evaluation set is split by classifier confidence into $X_n = \{x : f(x) < 0.2\}$ and $X_p = \{x : f(x) > 0.8\}$, up to $n_{\text{eval}} = 200$ each, and each directional policy generates from the corresponding group with its deterministic (mean) action.

Counterfactual Validity. We tighten the original, direction-agnostic definition to

$$\text{CV} = \Pr\left[\, |f(x_c) - f(x)| > \tau \wedge \text{sign}(f(x_c) - 0.5) = c \,\right], \tag{3}$$

with $\tau = 0.8$, so that an image moving the wrong way by a large margin is no longer silently credited.

FID. Computed on penultimate-layer features of a TorchXRayVision DenseNet-121 [4] rather than ImageNet Inception-v3, since it is trained on the same source distribution and yields a feature space sensitive to pneumonia-relevant structure rather than generic natural-image texture. Covariances are regularized ($\epsilon = 10^{-6}$) before the matrix square root, since the sample count is below the feature dimensionality.

Identity Preservation (IP). As a dependency-free substitute for FOP we report MS-SSIM (global anatomical identity across scales) and HF-corr, the Pearson correlation of high-pass-filtered images — a direct proxy for whether thin, high-contrast structures such as tubes, leads and pacemakers survive editing, since those are exactly the spatial frequencies a Gaussian high-pass isolates.

4.2. Ground-truth-anchored Lesion Hit Rate

CV, FID and IP share a blind spot: none of them can tell whether the edit landed on the pathology. RSNA ships 9,555 radiologist-drawn opacity boxes, which we retain. LHR measures the fraction of an edit's absolute-difference mass falling inside the annotation, against the box's own area as the null; *lift* (hit rate over null), not raw hit rate, is therefore the informative quantity. LHR is defined only for p→n, since n→p sources are classifier-negative and carry no annotation.

We calibrate the metric before use on synthetic edits with known ground truth: a uniform wash scores lift 1.00 ($p = 0.59$, chance, as it should), an on-lesion edit scores 12.4 ($p = 3 \times 10^{-49}$), an off-lesion edit scores 0.

4.3. Held-out transfer

Because CV is computed with the guidance classifier, it cannot separate a class-relevant edit from an adversarial perturbation. We therefore re-score every counterfactual with classifiers that never participated in guidance — `nih`, `pc`, `mimic_ch` — and report the ratio of their score shift to the guidance model's own. A ratio below 0.15 we treat as the adversarial regime. The denominator is read from each shard's own recorded configuration stamp rather than from the live configuration, so that a validator which never participated in guidance cannot dilute a denominator it never optimized.

4.4. Controls the comparison requires

A control-arm comparison is only as trustworthy as the plumbing beneath it, and three preconditions we had assumed did not hold. (i) A single dataset object was shared

by LoRA fine-tuning, the PPO pool and the metrics scan, so evaluation images had been seen during fine-tuning; we replace this with one deterministic, persisted, *patient-level* split shared by every stage, computed once because generation runs as independent array processes that cannot recompute an RNG-based split consistently. (ii) Training was unseeded — two identical submissions produced two different policies, survivable for a headline number but not for a comparison in which arms must differ by treatment alone. (iii) The control arms had never been exercised: wiring them up surfaced a routing bug under which every reduce silently fell back to the trained arm's shards, which would have made "the policy contributes nothing" an artefact of a missing flag, and a shared output directory under which each arm overwrote the previous one's metrics. A CPU-only invariant suite pins the LHR calibration, split disjointness and three further properties whose silent failure would yield a plausible-looking wrong number rather than a crash.

We report these because each was, at the time it was found, invisible in the outputs: all three would have produced a well-formed table.

5. Experiments

5.1. Setup

We use the RSNA Pneumonia Detection Challenge dataset [13] (DICOM, min–max normalized, resized to 256×256, binary label collapsed from bounding-box annotations). The guidance classifier is a TorchXRayVision DenseNet-121 (`densenet121-res224-rsna`); segmentation uses TorchXRayVision PSPNet for lung and heart channels; a YOLOv8n feature extractor supplies the foreign-object embedding. Each policy is trained for 800 episodes per direction; the two directions take 8h40m and 11h00m respectively on a single node.

5.2. Main results

Table 2 reports the committed configuration (2-model guidance ensemble, no Grad-CAM mask, no sparsity reward).

Three things are worth reading together. The policy succeeds at the base task in the easier direction and less so in the harder one. It concentrates edit mass on radiologist-annotated lesions far above chance — a $2.67\times$ lift is not a marginal effect. And yet it changes roughly 65% of pixels and places only about a quarter of its difference mass in the top 5% of the edit map, against a 0.05 uniform baseline and a ~ 0.5 threshold we would associate with a focal, lesion-shaped edit. The edits are class-informative and diffuse at the same time.

5.3. CV overstates what the edits achieve

Table 3 makes the case directly. Moving from single-classifier to 2-model ensemble guidance *lowers* mean CV from 0.965 to 0.838 while *raising* LHR lift from 2.293 to 2.673 and raising mean held-out transfer on both directions. A harder joint criterion is not a weaker policy, and the metric that fell is the one that shares a classifier with the generator.

A single-run point estimate cannot resolve the apparent `nih` regression, so we bootstrap it (10,000 percentile resamples of the $n = 200$ groups, numerator and denominator resampled jointly to preserve within-image pairing). For n→p, $[0.048, 0.114]$ vs. $[0.007, 0.084]$: the intervals overlap substantially and the apparent worsening is not distinguishable from noise — but both sit entirely below 0.15, so what *is* established is that n→p stays confidently adversarial under either policy. For p→n, $[0.076, 0.126]$ vs. $[0.155, 0.216]$ do not overlap, so that improvement is real. We report each policy's interval separately rather than a difference-of-ratios bootstrap, because the classifier that sorts images into X_n/X_p changed between runs and the two groups share only $\approx 70\%$ (n→p) and $\approx 24\%$ (p→n) of their members; a paired resample would synthesize a pairing that is not there.

This is the paper's central empirical claim. A system reporting CV 0.965 and, on the most independent validator, a transfer ratio of 0.08 is not producing explanations of the pathology; it is producing perturbations that a second model largely does not see. Nothing in the conventional metric suite surfaces that.

Table 2. Committed configuration, $n = 200$ per group. LHR is defined only for p→n.

Metric		**n→p**	**p→n**
CV	fraction valid, τ=0.8	0.960	0.715
	mean	0.838	
LHR lift	vs. annotation-area null	—	2.673
	t=24.19, p=4.2 × 10^{-53}, n=148		
Transfer	mean over 3 validators	0.282	0.293
EditConc.	top-5% mass (uniform=0.05)	0.253	0.318
	fraction of pixels changed	0.650	0.633
FID	vs. real target class	22.21	4.95
IP	MS-SSIM	0.923	0.870
IP	HF-corr	0.948	0.914

Table 3. Held-out transfer ratio, single- vs. ensemble-guided trained policies, $n = 200$ per group. Values below 0.15 are in the adversarial regime.

	nih		pc		mimic_ch	
Policy	n→p	p→n	n→p	p→n	n→p	p→n
Single-classifier	0.080	0.100	0.414	0.224	0.197	0.112
2-model ensemble	0.046	0.186	0.462	0.308	0.337	0.384

Table 4. Five trained-and-evaluated arms. Bold marks the arm no other arm beats under a paired test or bootstrap CI, not the best point estimate. *scored on the reduced leak-free validator set {nih,pc}; not comparable to the other rows' 3-validator mean.

Arm	LHR	Transfer	vs. default
Single-classifier	2.293	0.230/0.146	worse (d=0.482)
2-ens (default)	**2.673**	**0.282/0.293**	—
3-model ensemble	2.492	0.350/0.276*	unresolved (n=12)
Sparsity reward	2.398	0.311/0.212	worse (d=0.512)
Grad-CAM constraint	1.636	0.161/0.103	much worse (d=1.205)

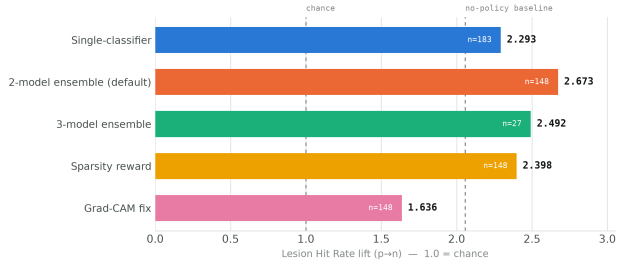


Figure 1. LHR lift (p→n) across all five trained arms. Dashed lines mark chance (1.0) and the no-policy label-protocol baseline (2.057).

5.4. Five trained arms

Every arm in Table 4 is a policy trained from scratch for 800 episodes per direction and evaluated end-to-end under the identical protocol — not a cheaper no-policy sweep or an offline proxy. Verdicts are paired significance tests on the images two arms share, never differences of group means.

Ensemble guidance helps, and the effect replicates. Paired per-image LHR-lift deltas across three independent seeds are $+0.0694$ ($d = 0.221$), $+0.0693$ ($d = 0.223$) and $+0.0575$ ($d = 0.151$), all with 95% CIs excluding zero. The single-classifier→2-model step is confirmed on the 76 shared annotated images ($d = 0.482$, $p = 7.1 \times 10^{-5}$). Growing the ensemble to three members improves `pc` n→p transfer from 0.462 to 0.623 with non-overlapping CIs, moves `nih` in the right direction on both counts without separating from noise, and does not resolve on LHR — only 12 annotated images are shared between the two arms, giving a paired CI of $[-0.298, +0.482]$. Ensemble size buys resistance to fooling an independent classifier, not sharper lesion targeting; the two properties dissociate.

The trained policy beats a random one, but barely. Against the random-action arm on the same 183 images: paired lift difference $+0.090$ (95% CI $[+0.021, +0.160]$), $t = 2.551$, $p = 0.0116$, Wilcoxon $p = 0.0147$, $d = 0.189$, trained better on 105/183. Real, small, and considerably smaller than either negative result below.

5.5. Two negative results

An annealed sparsity reward fails at its own objective. Edit concentration has stayed in the 0.25–0.32 band in every arm we measure. The sparsity term was built to address this and had been disabled throughout, leaving its annealing ramp inert. Enabled and isolated against the default, with $\lambda_{\text{sparsity}}$ confirmed ramping to its target, concentration moved by less than 0.02 and the fraction of pixels changed by less than 0.01. Meanwhile a paired test on the 148 shared annotated images gives a lift gap of $+0.275$ favouring the arm *without* it (95% CI $[+0.188, +0.363]$, $t = 6.229$, $p = 4.7 \times 10^{-9}$, $d = 0.512$), and p→n transfer falls from 0.293 to 0.212 with non-overlapping CIs on two of three validators. Without a working spatial mask the term degrades to plain unweighted L_1, which discourages edit mass uniformly rather than off-target mass specifically — a plausible account of why it competes with precise editing while failing at its own goal.

A Grad-CAM spatial constraint collapses the base task. Constraining edits to a Grad-CAM [15] region lowered LHR lift rather than raising it, so we diagnosed it directly. Scoring 263 member-image observations under the exact ensemble Grad-CAM path used during training contradicted the explanation on record: it is `chex`, not `rsna`, that degenerates (66/263, 25.1% vs. 1/263), and it does so concentrated at *low* predicted probability rather than near 0.5 — consistent with output saturation zeroing the gradient. When `chex` degenerates the averaged mask falls back toward the least-constrained member alone, and CAM–lesion overlap falls with it (2.052, $n = 127$ on clean episodes vs. 1.347, $n = 23$ on partially degenerate ones). The remedy on record — a finer target layer — does eliminate the degeneracy (0%/0.4%) but drops baseline CAM–lesion lift to 1.196, barely above chance.

Trained for real under that remedy, the result is unambiguous: CV collapses from 0.838 to 0.017, the paired LHR gap is $+1.037$ favouring the unconstrained arm (95% CI $[+0.900, +1.178]$, $t = 14.660$, $p = 1.5 \times 10^{-30}$, $d = 1.205$, better on 141/148 images), and transfer falls across all three validators. This is the largest paired effect size we measure anywhere, and it is against a mechanism introduced specifically to help. Constraining the action space to a finer, less confident CAM region does not trade a degeneracy problem for a localisation problem; it makes the base task substantially harder and every dependent metric degrades with it.

5.6. Clinical proxy metrics

We additionally report two clinically motivated proxies under the committed configuration: the Cardio-Thoracic Ratio (CTR), the ratio of heart-mask to lung-mask horizontal span, and a costophrenic-recess score (SCP) approximating pleural blunting, each tested with a paired t-test on the within-image real-to-counterfactual delta.

Table 5. Literature-reported reference values (Vanilla CycleGAN, GANterfactual [16]), not re-run under our protocol, vs. our committed configuration (measured, Table 2).

Method	CV	FID n→p	FID p→n
CycleGAN (lit.)	0.385	55.0	45.0
GANterfactual (lit.)	0.973	25.0	20.0
PPO-CF (ours, measured)	0.838	**22.21**	**4.95**

CTR shifts by $+0.0076$ ($t = 2.246$, $p = 0.026$) in n→p and by $+0.0080$ ($t = 1.010$, $p = 0.314$) in p→n: a small, only one-directionally significant shift, with an absolute magnitude (< 0.01) far below the metric's clinical resolution (its diagnostic threshold sits at 0.5). This is consistent with the intended behaviour of the background lock, since the cardiac silhouette lies largely outside the lung mask that gates editing — we do not read it as a confident, clinically directional CTR effect, only as a shift too small to act on either way. SCP's own computation flags itself as degenerate under this configuration (relative spread 0.19% across the four comparison groups), so its delta and t-test are withheld rather than reported on what is, numerically, a near-constant signal; built on a COCO-pretrained detector's feature norm rather than a dedicated recess detector, we do not consider it informative in its current form. Neither clinical proxy currently supports a directional claim about pleural or cardiac status; both should be read as evidence the edit is not clinically inert at best, not as measurements of a specific clinical change.

5.7. Comparison to prior methods

No Vanilla CycleGAN or GANterfactual training or evaluation artifact exists in our own pipeline, so Table 5 reports both as literature-reported reference values rather than same-protocol measurements: they were not re-run under §4's split, seeding, or evaluation procedure, and should be read as indicative context, not a controlled comparison. Relative to GANterfactual's reported FID, ours is better by $1.13\times$ (n→p, 22.21 vs. 25.0) and $4.04\times$ (p→n, 4.95 vs. 20.0); our mean CV (0.838) is lower than GANterfactual's reported 0.9725, which is expected under our protocol rather than a shortfall, since the committed configuration explicitly trades CV for the classifier-independent validity documented in §5.3 rather than optimizing CV in isolation.

5.8. Qualitative examples

Figure 2 shows representative counterfactuals from the committed configuration in both edit directions, generated with each policy's deterministic (mean) action. In both examples the classifier score fully saturates toward the target, well beyond the CV validity margin, and gross anatomical layout — rib cage, mediastinal silhouette, external hardware — is visibly retained outside the edited lung region: the qualitative signature of the background lock holding as intended. The p→n edit is visibly more concentrated than the n→p one, consistent with §3's directional asymmetry and with p→n's lower FID (Table 2), though both remain diffuse relative to a purely lesion-shaped edit (§5.4).

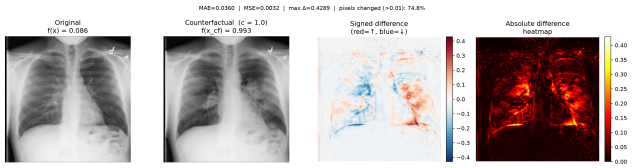


Figure 2. n→p. Original, counterfactual, signed difference, absolute difference. Change is concentrated over the lung fields where an opacity is synthesized, with anatomy outside the lung mask retained.

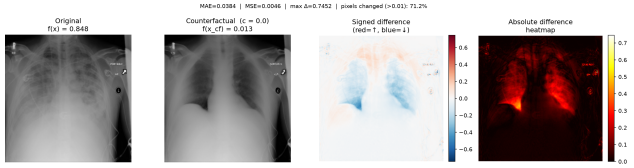


Figure 3. p→n. Original, counterfactual, signed difference, absolute difference. Change concentrates bilaterally over both lung fields, consistent with opacity removal, while gross anatomy and external hardware outside the lung mask are retained — the qualitative signature of the background lock holding.

6. Limitations

- **Edits remain diffuse.** No arm we trained moved edit concentration meaningfully; every one sits in the same 0.21–0.32 band regardless of what changed underneath it. Whether diffuseness is fixable at all within a gradient-guidance approach is open.
- **n→p stays adversarial.** On the most independent validator, both policies sit confidently below the 0.15 threshold. We can show ensemble guidance is a partial countermeasure; we cannot show the gap closes.
- **FOP is not computable offline**, so IP measures a related but not identical property, and SCP is a coarse proxy rather than a trained detector — and is, in the committed configuration, degenerate rather than merely coarse (§5.6).
- **Sample size limits several verdicts**, notably the 3-model ensemble's LHR comparison ($n = 12$ shared annotated images) and FID, whose variance at $n \sim 200$ in a 1024-d feature space is high.
- **Two independent policies** rather than one target-conditioned policy; the harder direction cannot borrow capacity from the easier one.

7. Conclusion

We presented a PPO-scheduled, LoRA-adapted latent diffusion pipeline for visual counterfactual explanation on chest radiographs, with a hard background lock confining edits to the lung field, and evaluated it under a protocol that pairs the field's conventional classifier-scored validity with a lesion-annotation-anchored hit rate and a held-out transfer ratio. The system attains a $2.67\times$ lesion-targeting lift over chance and improves transfer under ensemble guidance, replicated across seeds. It also shows, on its own outputs, that high CV coexists with diffuse edits and adversarial-regime transfer on the most independent validator — and that the configuration change which most improves the anchored metrics is the one that *lowers* CV. Two mechanisms designed to concentrate edits were trained and measured rather than assumed, and both are confirmed harmful, one by the largest effect size in the study. We take the broader implication to be methodological: CV computed with the guidance classifier should not be reported as a standalone success criterion, and an anchored companion metric — to human annotation, to model independence, or both — costs little and changes the conclusion.

References

[1] Maximilian Augustin, Valentyn Boreiko, Francesco Croce, and Matthias Hein. Diffusion visual counterfactual explanations. In *NeurIPS*, 2022. 2

[2] Omri Avrahami, Dani Lischinski, and Ohad Fried. Blended diffusion for text-driven editing of natural images. In *CVPR*, 2022. 2

[3] Kevin Black, Michael Janner, Yilun Du, Ilya Kostrikov, and Sergey Levine. Training diffusion models with reinforcement learning. In *ICLR*, 2024. 2

[4] Joseph Paul Cohen, Joseph D. Viviano, Paul Bertin, Paul Morrison, Parsa Torabian, Matteo Guarrera, Matthew P. Lungren, Akshay Chaudhari, Rupert Brooks, Mohammad Hashir, and Hadrien Bertrand. TorchXRayVision: A library of chest x-ray datasets and models. In *MIDL*, 2022. 3

[5] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat GANs on image synthesis. In *NeurIPS*, 2021. 2

[6] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *ICLR*, 2022. 2

[7] Guillaume Jeanneret, Loïc Simon, and Frédéric Jurie. Diffusion models for counterfactual explanations. In *ACCV*, 2022. 2

[8] Guillaume Jeanneret, Loïc Simon, and Frédéric Jurie. Adversarial counterfactual visual explanations. In *CVPR*, 2023. 2

[9] Andreas Lugmayr, Martin Danelljan, Andres Romero, Fisher Yu, Radu Timofte, and Luc Van Gool. RePaint: Inpainting using denoising diffusion probabilistic models. In *CVPR*, 2022. 2

[10] Chenlin Meng, Yutong He, Yang Song, Jiaming Song, Jiajun Wu, Jun-Yan Zhu, and Stefano Ermon. SDEdit: Guided image synthesis and editing with stochastic differential equations. In *ICLR*, 2022. 2

[11] Konpat Preechakul, Nattanat Chatthee, Suttisak Wizadwongsa, and Supasorn Suwajanakorn. Diffusion autoencoders: Toward a meaningful and decodable representation. In *CVPR*, 2022. 2

[12] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, 2022. 2

[13] RSNA. RSNA pneumonia detection challenge, 2018. Kaggle. https://www.kaggle.com/c/rsna-pneumonia-detection-challenge. 4

[14] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. 2

[15] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *ICCV*, 2017. 5

[16] Sumedha Singla, Motahhare Eslami, Brian Pollack, Stephen Wallace, and Kayhan Batmanghelich. Explaining the black-box smoothly – a counterfactual approach. *Medical Image Analysis*, 2022. 2, 3, 6

[17] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *ICLR*, 2021. 2